

Assignment 1: Bitcoin Blockchain Exploration

Part A: One-Word Answerable Questions from bitcoin.org

1. Bitcoin was introduced by whom?

Satoshi Nakamoto

2. Bitcoin transactions are recorded on which data structure?

Blockchain

3. Bitcoin uses which consensus mechanism?

Proof-of-Work

4. The smallest unit of Bitcoin is called?

Satoshi

5. Bitcoin is a peer-to-peer type of what system?

Cash

6. A wallet stores the user's _____ key.

Private

7. The recovery phrase used in wallets is called a _____ phrase.

Seed

8. Hardware wallets store keys _____ (online/offline).

Offline

9. Bitcoin addresses are derived from _____ keys.

Public

10. What is required to sign a transaction? (public/private)

Private key

11. Bitcoin prevents double-spending using _____.

Consensus

12. Bitcoin is secured using _____ cryptography (asymmetric/symmetric).

Asymmetric

13. To run a Bitcoin node, the software required is Bitcoin _____.

Core

14. The process of verifying transactions is called _____.

Validation

15. The Bitcoin network is described as _____ (centralized/decentralized).

Decentralized

16. Miners solve _____ puzzles to add blocks.

Hash

17. Mining rewards a block with newly created _____.

Bitcoin

18. The block time target for Bitcoin is _____ minutes.

10

19. Mining difficulty adjusts every _____ blocks.

2016

20. Miners compete to find a valid block _____ (hash/signature).

Hash

21. Each block contains a _____ root.

Merkle

22. Blocks are linked using the previous block's _____.

Hash

23. A Bitcoin transaction contains inputs and _____.

Outputs

24. Unspent transaction outputs are called _____.

UTXO

25. The first-ever Bitcoin block is called the _____ block.

Genesis

26. A full node stores the entire blockchain _____.

History

27. Bitcoin uses which protocol for communication? (TCP/P2P)

P2P

28. A light wallet uses _____ mode (SPV/full).

SPV

29. Nodes relay information across a _____ network.

Peer-to-peer

30. The maximum Bitcoin supply is _____ million.

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Task B1: Explore a Block

Block height:

961,792

Block hash:

[illegible]

Timestamp:

10 August 2026, approximately 08:37 UTC

Number of transactions:

6,020

Miner/pool name:

Not available in retrieved block information

Task B2: Explore a Transaction**10. Write the TXID of the transaction.**

e28dc8962e665e77a9c496f4593ccab738e6170babb14e240012cfa9706fdf06

11. Number of inputs:

2

12. Number of outputs:

2

13. Transaction fee:

465 satoshis (0.00000465 BTC)

14. How many confirmations does it have?

Approximately 3,396 confirmations

Task B3: Explore an Address**15. Search for any Bitcoin address shown in the block.**

bc1q4k3cz4knyu628d3njxntmt59a4aj9dwafyyjw

Total BTC received:

0.00649225 BTC

Total BTC sent:

0.01602434 BTC

Current balance:

Requires complete address history; not determined from one transaction.

What is the role of a full node?

A full node validates all transactions and blocks, stores the blockchain, rejects invalid data and supports decentralization.

List any two developer resources available on bitcoin.org.

Bitcoin Developer Documentation and Bitcoin Developer Guides.

Task C1: Market Information**Current price of BTC:**

\$65,108.96 USD

24-hour % change:

+0.55%

Market capitalization:

Approximately \$1.30 trillion USD

Circulating supply:

Approximately 20.06 million BTC

Total/maximum supply:

Maximum supply: 21 million BTC

Task C2: Historical Data**What was the price of Bitcoin one year ago?**

\$119,306.76 USD approximately

Highest price in last 30 days:

\$66,923.95 USD approximately

Lowest price in last 30 days:

\$61,765 USD approximately

Investopedia Exploration**Why is Bitcoin considered volatile?**

Bitcoin is volatile because of rapid changes in demand, market sentiment, regulations, speculation and adoption.

List two investment risks mentioned.

1. Price volatility
2. Regulatory uncertainty

Part E: Reflection & Analysis**Which website gave you the most clear understanding? Why?**

bitcoin.org provided the clearest understanding because it gives official information about Bitcoin concepts, wallets, mining and security.

Explain how blockchain ensures trust without a central authority.

Blockchain uses a distributed ledger maintained by many nodes. Transactions are verified through cryptography and consensus mechanisms. Blocks are connected using hashes, making alteration difficult. This allows users to trust the network without a central authority.